

See the Emotion: A Facial Emoji Proxy Modeling for EEG Emotion Recognition

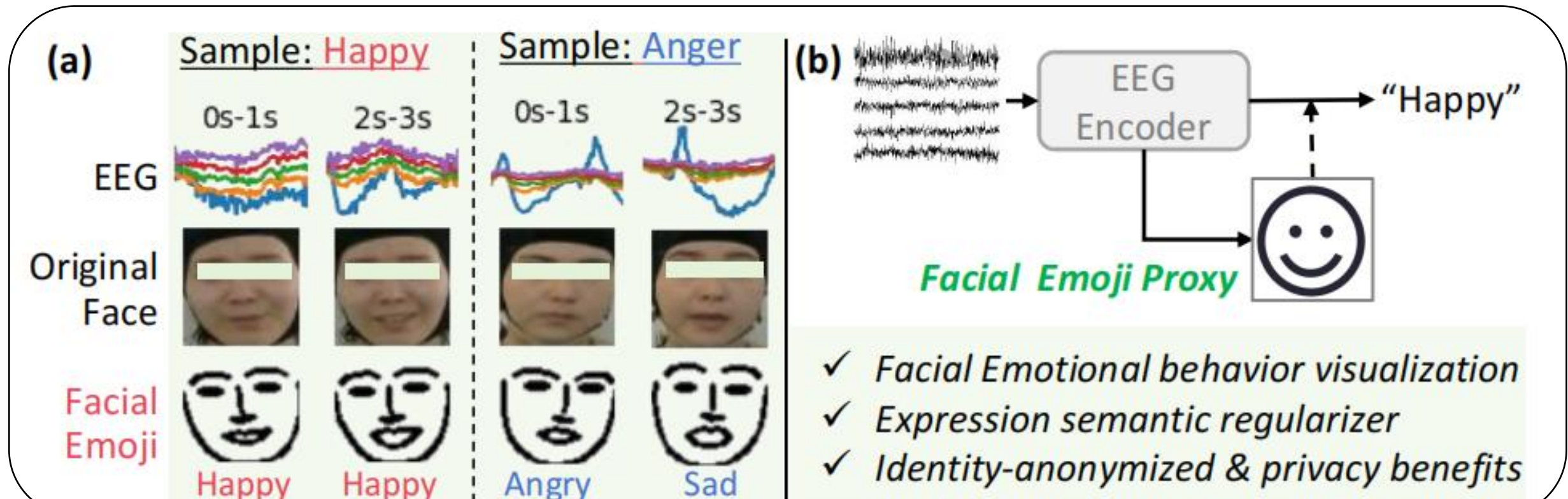
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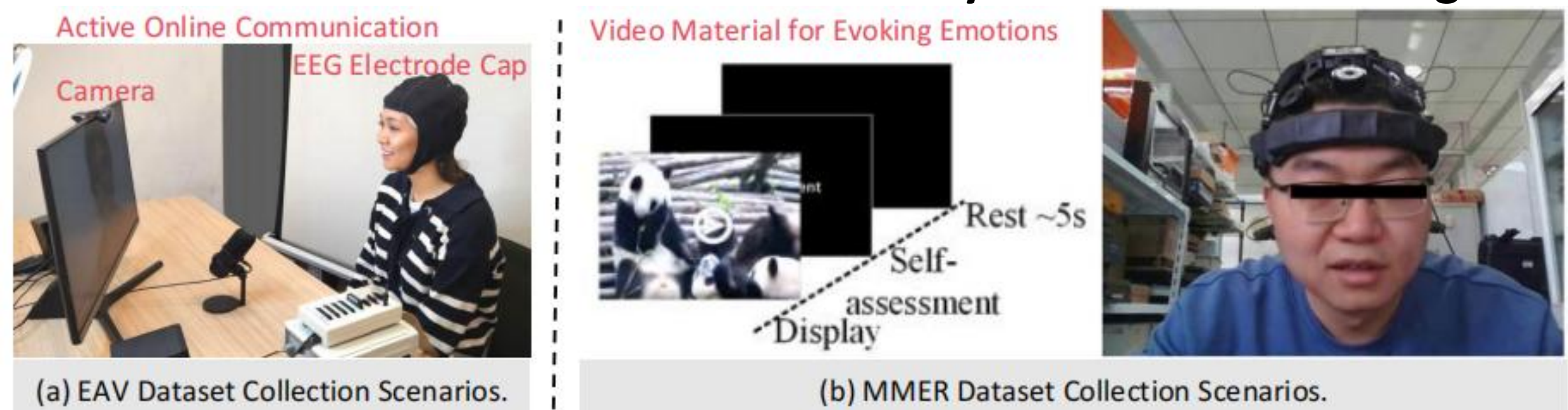


1. Introduction

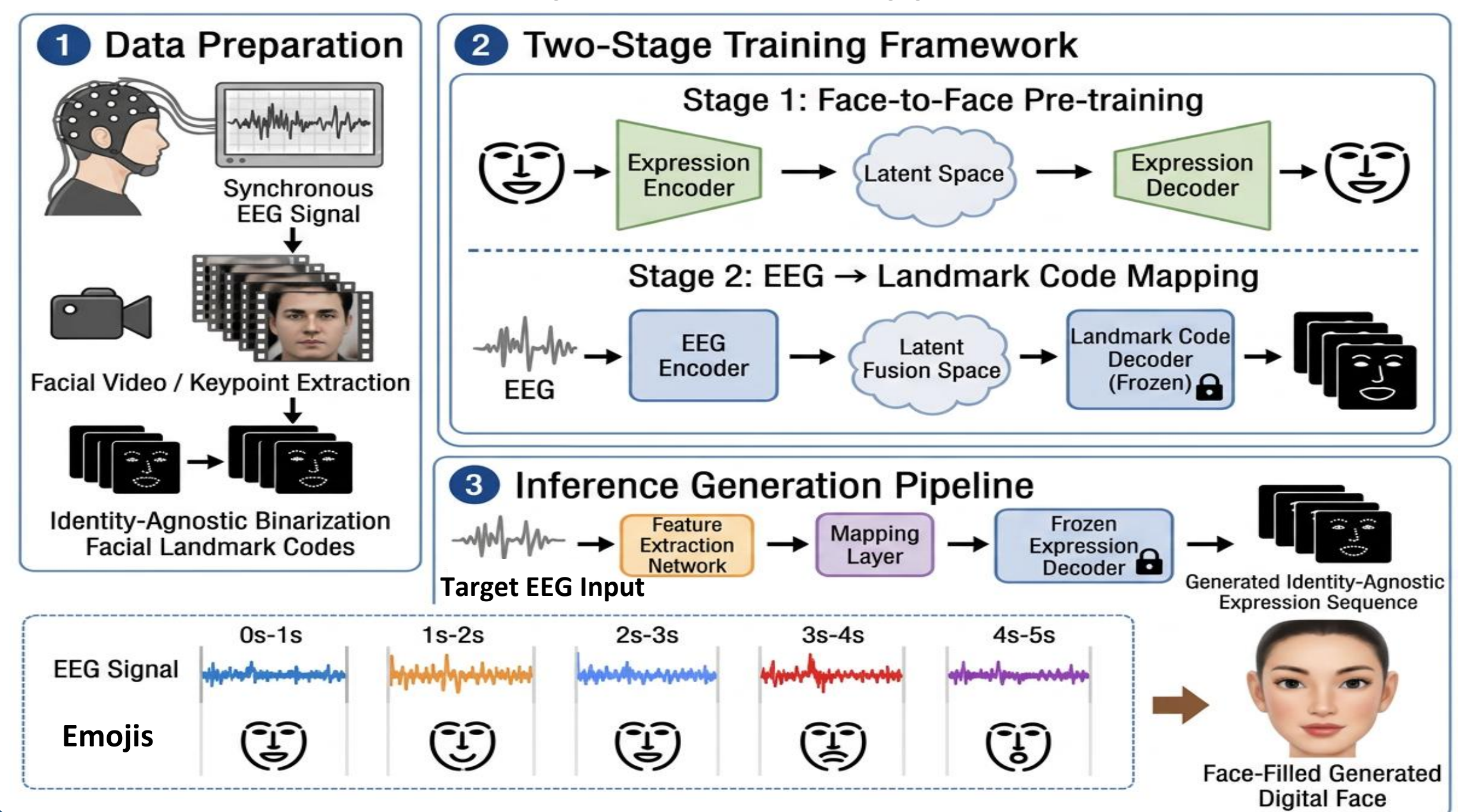
- **Challenge:** Despite the high accuracy of EEG-based emotion recognition, existing models remain opaque “black boxes”, lacking semantic grounding between abstract neural features and human interpretable states.
- **What we do:** we reframe EEG explainability as a cross-modal generation task, shifting the paradigm from feature attribution to behavioral visualization. We introduce Facial Emoji Proxy Modeling, a novel framework that translates high-dimensional EEG signals into identity-anonymized facial emojis.
- **Motivation:** Guided by the neuroscientific inspiration of neural-facial association, this approach grounds neural representations in the manifold of observable facial dynamics.



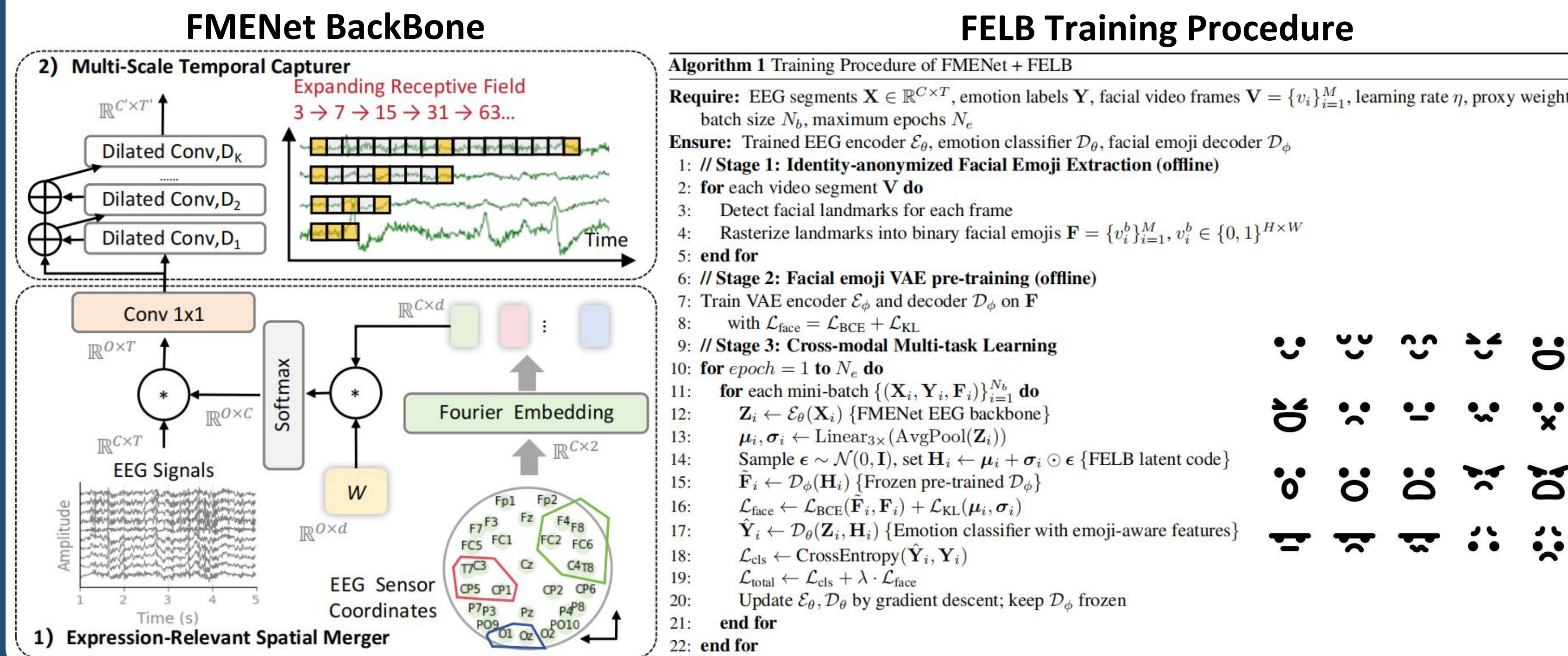
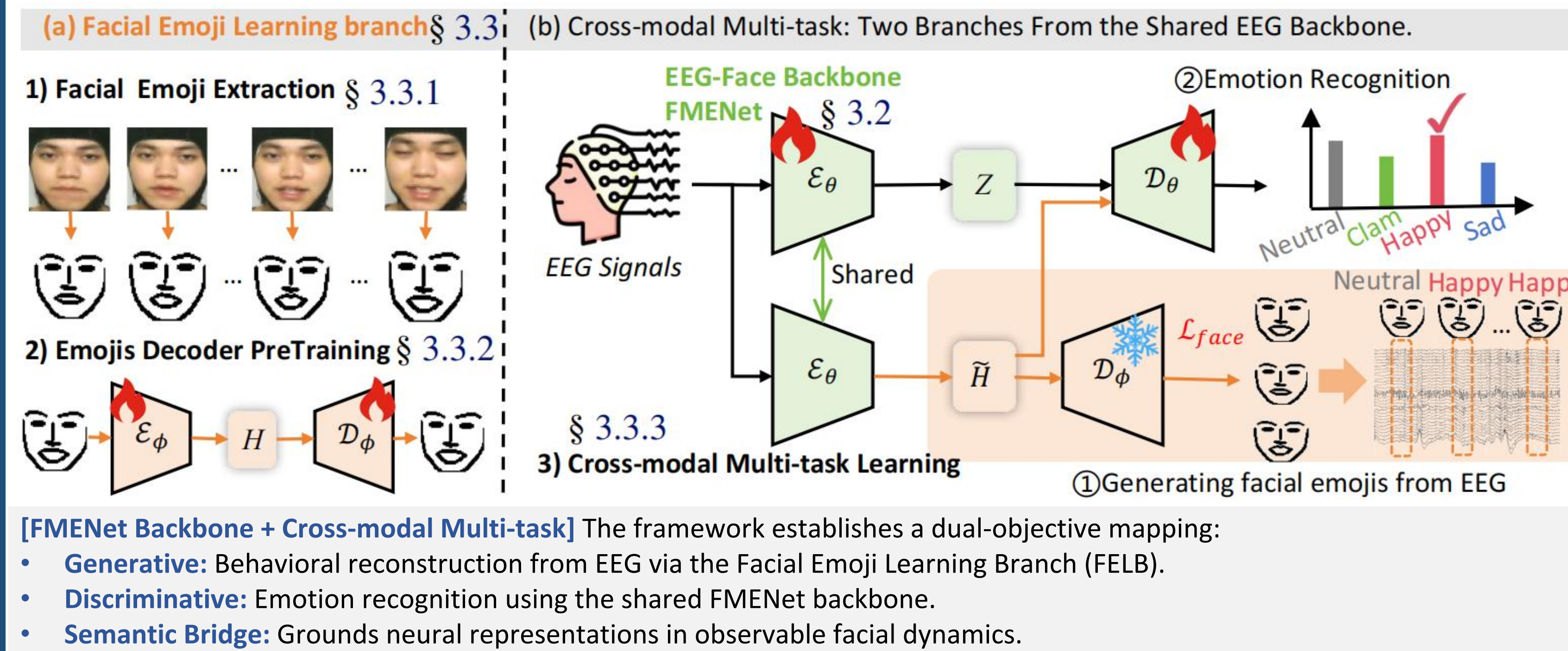
Two recent EEG-face datasets with synchronized recordings



Overview of the Entire Pipeline, Future Applications of Virtual Faces



2. Method: Facial Emoji Proxy Modeling



3. Main Results

Emotion Recognition (EEG-only Inference)

Method	Ref.	Train Modality	Infer. Modality	Accuracy↑	F1-score↑
EEG-Only Models					
EEGNet (Lawhern et al., 2018)	JNE'18	EEG	EEG	60.00	58.00
EEGformer (Wan et al., 2023)	Front. Neurosci.'23	EEG	EEG	53.50	52.00
Hyper-MML(E) (Kang et al., 2025)	arXiv'25	EEG	EEG	71.83	72.04
FMENet	Ours	EEG	EEG	76.96	76.47
Multimodal Models (EEG+Face+Audio)					
bc-LSTM (Ma et al., 2019)	ACM MM'19	EEG+Face+Audio	EEG+Face+Audio	57.21	57.30
DialogueRNN (Majumder et al., 2019)	AAAI'19	EEG+Face+Audio	EEG+Face+Audio	61.20	61.14
M3NET (Mane et al., 2020)	EMBC'20	EEG+Face+Audio	EEG+Face+Audio	75.14	75.42
HAUL (Yi et al., 2024)	ACM MM'24	EEG+Face+Audio	EEG+Face+Audio	75.91	75.92
AGF-IB (Shou et al., 2024)	Inf. Fusion'24	EEG+Face+Audio	EEG+Face+Audio	76.19	76.08
Hyper-MML(EFA) (Kang et al., 2025)	arXiv'25	EEG+Face+Audio	EEG+Face+Audio	78.21	77.80
Ours (EEG-Only Inference with Emoji Explanation)					
FMENet+FELB	Ours	EEG+Face	EEG	77.13	77.02

User Study

Accuracy

Mean = 0.76

0.72

0.76

0.80

0.78

0.70

0.72

0.66

0.82

0.84

0.78

P1

P2

P3

P4

P5

P6

P7

P8

P9

P10

Participant

Calm

0.75

Happiness

0.87

Anger

0.76

Sadness

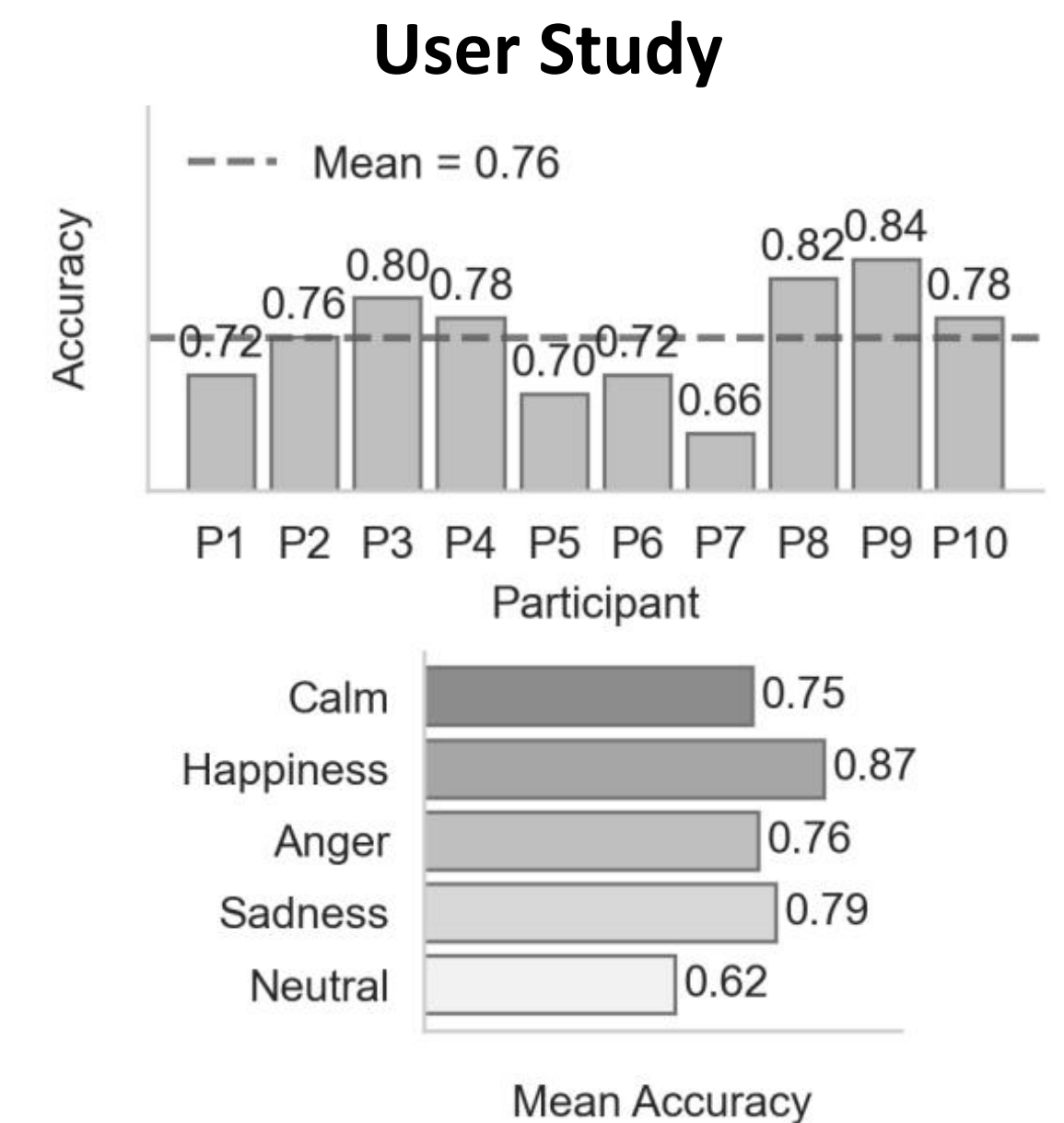
0.79

Neutral

0.62

Mean Accuracy

Non-expert users reliably interpret the EEG-reconstructed emoji clips with high accuracy



Non-expert users reliably interpret the EEG-reconstructed emoji clips with high accuracy

4. More Analysis

Analysis of Computational Cost for Deployment Potential

Model	#Params (M) ↓	FLOPs (GLOPs) ↓	Train Time	Infer. Speed (samples/s)	Acc. (EAV, %) ↑
EEGNet	0.005	0.014	~4h	212.6 (Cls.)	58.99
EEGNet+Our FELB	0.005 (+0.001)	0.014 (+0.001)	~4.5h	208.3/52.1 (Cls.+Gen.)	56.35
EEGConformer	0.454	1.363	~5h	46.6 (Cls.)	63.33
EEGConformer+Our FELB	0.454 (+0.001)	1.363 (+0.001)	~5.5h	45.8/11.5 (Cls.+Gen.)	62.75
FMENet	0.132	1.95	~5h	48.2 (Cls.)	76.96
FMENet+FELB (Our full model)	0.132 (+0.001)	1.95 (+0.001)	~6h	47.9/12.0 (Cls.+Gen.)	77.13

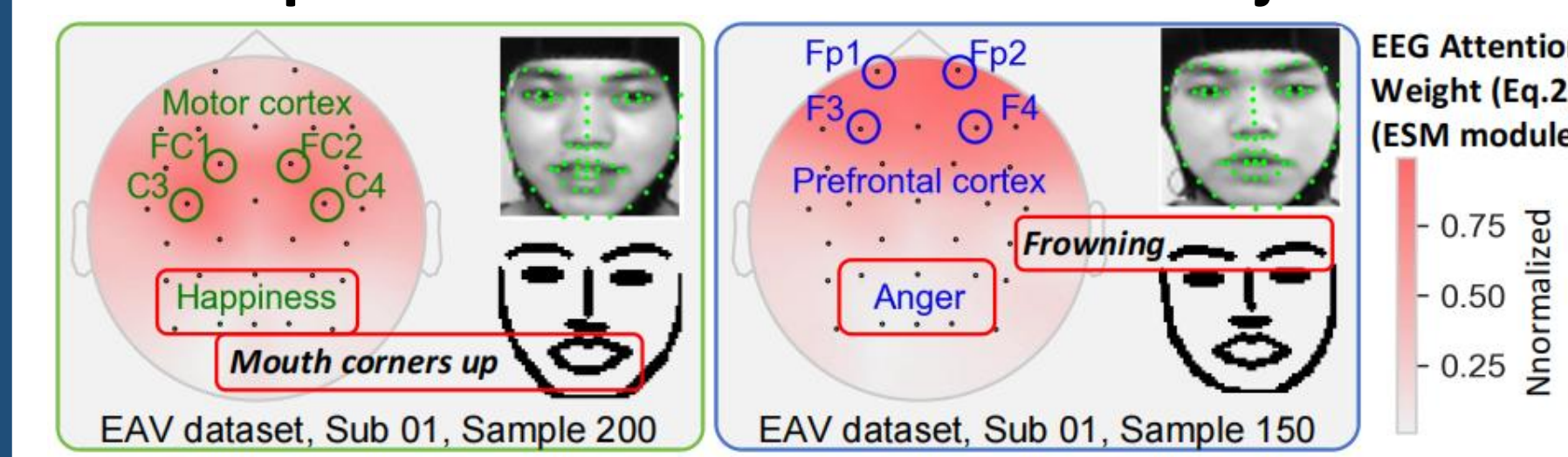
Ablation of key components

Method	Acc.↑	F1↑	MMER	Acc.↑	F1↑
FMENet + FELB (full model)	77.13	77.02	69.73	64.28	
w/o FELB (§ 3.3)	76.96	76.47	69.11	63.16	
w/o ESM module (§ 3.2.1)	65.87	65.29	59.62	55.78	
w/o MEC module (§ 3.2.2)	68.34	67.58	60.73	56.30	

Zero-shot on SEED

Metric	Positive	Neutral	Negative	Avg.
Emoji Acc. (%)	72.31	58.37	49.58	60.09
Label Cons. (%)	74.15	60.23	51.07	61.82

EEG spatial-attention drivers for emoji actions

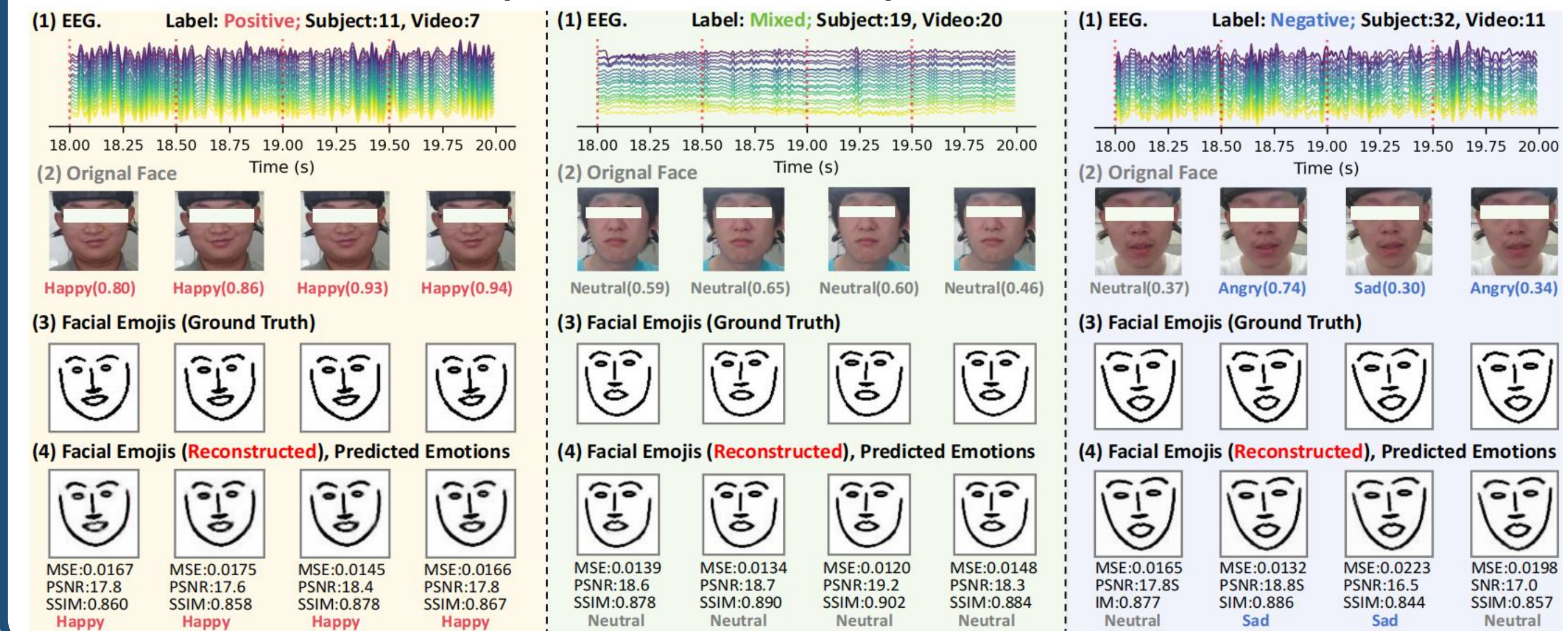


ESM (§3.2.1) spatial attention on the scalp and shows that the motor cortex is emphasized when the reconstructed emoji exhibits mouth corner up (e.g., smiling), whereas the prefrontal cortex receives higher attention during brow-related negative expressions.

Brain region ablation via spatial attention masking

Masked Region	Calm	Happiness	Anger	Sadness	Neutral	Avg.
None (Full)	0.9105	0.9510	0.9020	0.8860	0.9195	0.9138
Motor cortex	0.8895	0.7912	0.8728	0.8791	0.9023	0.8670
Prefrontal	0.8714	0.9108	0.7426	0.7639	0.8817	0.8341

Visualization of representative examples from the MMR dataset



5. Information

Paper

Code

CV

